

HDOC DAY-23

Title: Square and Rectangular Pattern Generation using Parameterless Functions and Nested Loops

1. Problem Statement

Write a menu-driven C program using a switch statement that allows users to generate basic square and rectangular patterns using nested loops without arrays. Each pattern activity must be implemented in a separate parameterless function returning no value (void):

1. **Solid square pattern:** Input n , print an $n \times n$ grid of *.
2. **Solid rectangular pattern:** Input m and n , print an $m \times n$ grid of *.
3. **Row-number pattern:** Input m and n , print an $m \times n$ grid where elements equal row number.
4. **Column-number pattern:** Input m and n , print an $m \times n$ grid where elements equal column number.
5. **Exit:** Terminate execution.

2. Algorithms

Main Control Flow

1. Display the option menu inside a do-while loop.

2. Read user choice and execute the respective parameterless function using a switch statement.
3. Terminate loop when option 5 is selected.

Pattern Functions

- **print_solid_square():** Read n . Outer loop i from 0 to $n - 1$, inner loop j from 0 to $n - 1$: print *.
- **print_solid_rectangle():** Read m and n . Outer loop i from 0 to $m - 1$, inner loop j from 0 to $n - 1$: print *.
- **print_row_number_pattern():** Read m and n . Outer loop i from 1 to m , inner loop j from 1 to n : print i .
- **print_column_number_pattern():** Read m and n . Outer loop i from 1 to m , inner loop j from 1 to n : print j .

3. Test Case Table

Test Case	Pattern / Option	Input Parameters	Expected Pattern Output	Actual Output	Status
1	Solid Square	$n = 4$	4×4 grid of *	Matched	Pass

Test Case	Pattern / Option	Input Parameters	Expected Pattern Output	Actual Output	Status
				expected grid	
2	Solid Rectangle	$m = 3, n = 5$	3×5 grid of *	Matched expected grid	Pass
3	Row-Number Pattern	$m = 4, n = 4$	4 rows filled with numbers 1 to 4	Matched expected grid	Pass
4	Column-Number Pattern	$m = 4, n = 4$	4 rows repeating column	Matched expected grid	Pass

Test Case	Pattern / Option	Input Parameters	Expected Pattern Output	Actual Output	Status
			s 1 2 3 4		
5	Invalid Choice	Choice = 9	"Error: Invalid choice!" "message"	Error message shown	Pass
6	Exit Option	Choice = 5	Terminate loop and program	Program exited	Pass

4. Source Code Listing

```
#include <stdio.h>

void print_solid_square(void) {
    int n;
    printf("\n--- Solid Square Pattern ---\n");
```

```

printf("Enter number of rows/columns (n): ");
scanf("%d", &n);

if (n <= 0) {
    printf("Error: Input must be a positive integer.\n");
    return;
}

for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        printf("*");
    }
    printf("\n");
}
}

void print_solid_rectangle(void) {
    int m, n;
    printf("\n--- Solid Rectangular Pattern ---\n");
    printf("Enter number of rows (m): ");
    scanf("%d", &m);
    printf("Enter number of columns (n): ");
    scanf("%d", &n);

    if (m <= 0 || n <= 0) {
        printf("Error: Rows and columns must be positive integers.\n");
        return;
    }

    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            printf("*");
        }
        printf("\n");
    }
}

void print_row_number_pattern(void) {
    int m, n;
    printf("\n--- Row-Number Pattern ---\n");

```

```

printf("Enter number of rows (m): ");
scanf("%d", &m);
printf("Enter number of columns (n): ");
scanf("%d", &n);

if (m <= 0 || n <= 0) {
    printf("Error: Rows and columns must be positive integers.\n");
    return;
}

for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        printf("%d ", i);
    }
    printf("\n");
}
}

void print_column_number_pattern(void) {
    int m, n;
    printf("\n--- Column-Number Pattern ---\n");
    printf("Enter number of rows (m): ");
    scanf("%d", &m);
    printf("Enter number of columns (n): ");
    scanf("%d", &n);

    if (m <= 0 || n <= 0) {
        printf("Error: Rows and columns must be positive integers.\n");
        return;
    }

    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
}

int main(void) {

```

```
int choice;

do {
    printf("\n===== PATTERN MENU
=====\\n");
    printf("1. Solid square pattern\\n");
    printf("2. Solid rectangular pattern\\n");
    printf("3. Row-number pattern\\n");
    printf("4. Column-number pattern\\n");
    printf("5. Exit\\n");
    printf("Enter choice (1-5): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            print_solid_square();
            break;
        case 2:
            print_solid_rectangle();
            break;
        case 3:
            print_row_number_pattern();
            break;
        case 4:
            print_column_number_pattern();
            break;
        case 5:
            printf("Exiting program...\\n");
            break;
        default:
            printf("Error: Invalid choice! Please enter an option between 1 and
5.\\n");
            break;
    }
} while (choice != 5);

return 0;
}
```

4. Execution & Verification Screenshots

```
Session Actions Edit View Help
(kali@kali)-[~]
└─$ mkdir HDOC_Day23
cd HDOC_Day23
(kali@kali)-[~/HDOC_Day23]
└─$ nano day23.c
(kali@kali)-[~/HDOC_Day23]
└─$ gcc day23.c -o day23
(kali@kali)-[~/HDOC_Day23]
└─$ ./day23
File System
===== PATTERN MENU =====
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter choice (1-5): 1

— Solid Square Pattern —
Enter number of rows/columns (n): 4
****
****
****
****

===== PATTERN MENU =====
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter choice (1-5): 2

— Solid Rectangular Pattern —
Enter number of rows (m): 3
Enter number of columns (n): 5
*****
*****
*****

===== PATTERN MENU =====
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter choice (1-5): 3

— Row-Number Pattern —
Enter number of rows (m): 4
Enter number of columns (n): 4
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

===== PATTERN MENU =====
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter choice (1-5): 4

— Column-Number Pattern —
Enter number of rows (m): 4
Enter number of columns (n): 4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

===== PATTERN MENU =====
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter choice (1-5): 5
Exiting program ...
(kali@kali)-[~/HDOC_Day23]
```